



=== Harwell RSX Characterization Report ===

Each resistor undergoes characterization at multiple temperature points to create a precise model for calculating resistance across its operating range. Our model for resistors use a proprietary metal foil design that depending on the unique assembly of each resistor, we may characterize it via a polynomial or third-degree fit.

=== Instrument Summary ===

Date: September 15, 2026

Model: Harwell RS-7

Value: 100 Ω

S/N: RS70271

Public Report Reference: 6AAB0A85

=== Calibration Data ===

Calibrated Resistance: 100.001540 Ω at 23°C

Expanded Uncertainty of Measurement, k=2: 0.8 PPM

Measurement Current: 1 mA

=== Temperature Characterization Data ===

α : 1.382E-07 $\Omega/\Omega/^\circ\text{C}$ (0.138 ppm/ $^\circ\text{C}$)

β : 1.451E-08 $\Omega/\Omega/^\circ\text{C}^2$ (0.015 ppm/ $^\circ\text{C}^2$)

Linear TCR (18-28°C): 0.14 ppm/ $^\circ\text{C}$

To calculate the resistor value at a temperature other than 23°C, use the following model where T is the desired temperature in $^\circ\text{C}$:

$$R(T) = R_{(23)} [1 + \alpha(T - 23) + \beta(T - 23)^2]$$

Note: This model may not be accurate outside the characterized temperature range.

Temperature ($^\circ\text{C}$)	Temperature ($^\circ\text{F}$)	Resistance (Ω)	Deviation (ppm)
18.0	64.4	100.001507	-0.33
18.5	65.3	100.001507	-0.33
19.0	66.2	100.001508	-0.32
19.5	67.1	100.001509	-0.31
20.0	68.0	100.001512	-0.28
20.5	68.9	100.001515	-0.25
21.0	69.8	100.001518	-0.22
21.5	70.7	100.001523	-0.17
22.0	71.6	100.001528	-0.12
22.5	72.5	100.001533	-0.07
23.0	73.4	100.001540	0.00
23.5	74.3	100.001547	0.07
24.0	75.2	100.001555	0.15
24.5	76.1	100.001564	0.24
25.0	77.0	100.001573	0.33
25.5	77.9	100.001584	0.44
26.0	78.8	100.001595	0.55
26.5	79.7	100.001606	0.66
27.0	80.6	100.001618	0.78
27.5	81.5	100.001632	0.92
28.0	82.4	100.001645	1.05

Harwell Resistor TCR
Model: RS-7 | Nominal Value: 100 Ω | Serial Number: RS70271

